

Customer Purchase Pattern Analyzer

Executive Summary

This project analyzes customer purchase behavior, revenue drivers, customer segments, and retention opportunities using a business intelligence dashboard.

Business Problem

Organizations need to understand which customers, products, and regions drive revenue and where retention efforts should be focused.

Dataset Overview

Dataset includes customer demographics, purchases, payment methods, loyalty status, product categories, and regional information.

Data Cleaning

Handled missing values, removed duplicates, standardized names and categories, validated dates, and checked revenue calculations.

Feature Engineering

Created customer segments, age groups, revenue bands, purchase frequency indicators, and basic CLV measures.

KPIs

Revenue ■25.68M, Customers 1,248, AOV ■2,365, Repeat Purchase Rate 42.75%, CLV ■8,943.

Dashboard Analysis

Analyzed revenue trends, product performance, regional performance, top customers, payment methods, and customer age groups.

Key Insights

VIP customers generate the highest revenue. Electronics is the best-performing category. North region leads revenue. UPI is the preferred payment method.

Business Recommendations

Expand loyalty programs, run retention campaigns, promote electronics bundles, and improve regional marketing.

Conclusion

The project demonstrates practical customer analytics and business intelligence skills applicable to retail and e-commerce environments.